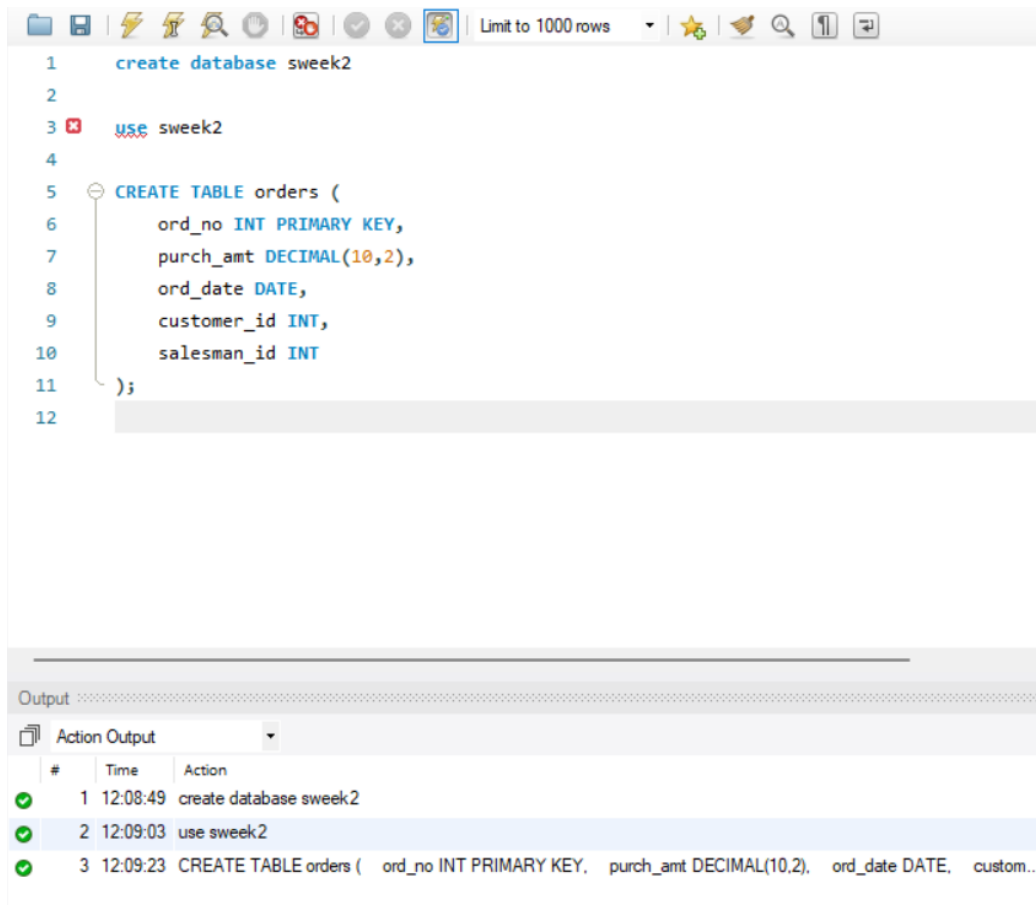


Skill week-2

CREATING TABLE



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search. The main editor area contains the following SQL code:

```
1 create database sweek2
2
3 use sweek2
4
5 CREATE TABLE orders (
6     ord_no INT PRIMARY KEY,
7     purch_amt DECIMAL(10,2),
8     ord_date DATE,
9     customer_id INT,
10    salesman_id INT
11 );
12
```

Below the editor is an 'Output' section with a dropdown menu set to 'Action Output'. It displays a log of executed actions:

#	Time	Action
✓ 1	12:08:49	create database sweek2
✓ 2	12:09:03	use sweek2
✓ 3	12:09:23	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, custom...

INSERTING DATA

```

13
14 • INSERT INTO orders VALUES
15 (70001,150.50,'2012-10-05',3005,5002),
16 (70009,270.65,'2012-09-10',3001,5005),
17 (70002,65.26,'2012-10-05',3002,5001),
18 (70004,110.50,'2012-08-17',3009,5003),
19 (70007,948.50,'2012-09-10',3005,5002),
20 (70005,2400.60,'2012-07-27',3007,5001),
21 (70008,5760.00,'2012-09-10',3002,5001),
22 (70010,1983.43,'2012-10-10',3004,5006),
23 (70003,2480.40,'2012-10-10',3009,5003),
24 (70012,250.45,'2012-06-27',3008,5002);
25

```

Output

Action Output

#	Time	Action
✓ 1	12:08:49	create database week2
✓ 2	12:09:03	use week2
✓ 3	12:09:23	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, custo...
✓ 4	12:12:34	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005), (70...

WHERE CLAUSE

```

25
26 • SELECT *
27 FROM orders
28 WHERE purch_amt > 2000;
29
30

```

Result Grid

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
*	NULL	NULL	NULL	NULL	NULL

orders 1 x

Output

Action Output

#	Time	Action
✓ 1	12:08:49	create database week2
✓ 2	12:09:03	use week2
✓ 3	12:09:23	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, custo...
✓ 4	12:12:34	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005), (70...
✓ 5	12:13:53	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000

```

30 • SELECT *
31 FROM orders
32 WHERE ord_date='2012-09-10';
33

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
* NULL	NULL	NULL	NULL	NULL

orders 2 x

Output

Action Output

#	Time	Action
✓ 1	12:08:49	create database sweek2
✓ 2	12:09:03	use sweek2
✓ 3	12:09:23	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, custo...
✓ 4	12:12:34	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005), (70...
✓ 5	12:13:53	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000
✓ 6	12:14:21	SELECT * FROM orders WHERE ord_date='2012-09-10' LIMIT 0, 1000

```

34 • SELECT *
35 FROM orders
36 WHERE salesman_id=5001;
37

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
* NULL	NULL	NULL	NULL	NULL

orders 3 x

Output

Action Output

#	Time	Action
✓ 1	12:08:49	create database sweek2
✓ 2	12:09:03	use sweek2
✓ 3	12:09:23	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, custo...
✓ 4	12:12:34	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005), (70...
✓ 5	12:13:53	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000
✓ 6	12:14:21	SELECT * FROM orders WHERE ord_date='2012-09-10' LIMIT 0, 1000
✓ 7	12:14:51	SELECT * FROM orders WHERE salesman_id=5001 LIMIT 0, 1000

ORDER BY

```

38 • SELECT *
39 FROM orders
40 ORDER BY purch_amt DESC;
41

```

```

42 • SELECT *
43 FROM orders
44 ORDER BY ord_date;
45

```

Result Grid

Filter Rows:

Edit:

Export/Import:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70008	5760.00	2012-09-10	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70010	1983.43	2012-10-10	3004	5006
	70007	948.50	2012-09-10	3005	5002
	70009	270.65	2012-09-10	3001	5005

Result Grid

Filter Rows:

Edit:

Export/Import:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70012	250.45	2012-06-27	3008	5002
	70005	2400.60	2012-07-27	3007	5001
	70004	110.50	2012-08-17	3009	5003
	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005

SUM()

```
45
46 • SELECT SUM(purch_amt) AS total_revenue
47 FROM orders;
48
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_revenue			
14420.29			

Result 6 x

Output

 Action Output

#	Time	Action
	4 12:12:34	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-10-05',3005,5002)
	5 12:13:53	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000
	6 12:14:21	SELECT * FROM orders WHERE ord_date='2012-09-10' LIMIT 0, 1000
	7 12:14:51	SELECT * FROM orders WHERE salesman_id=5001 LIMIT 0, 1000
	8 12:15:39	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000
	9 12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
	10 12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000

AVG()

```

49 • SELECT AVG(purch_amt) AS average_order
50 FROM orders;
51

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	average_order			
▶	1442.029000			

Result 7 ×

Output

Action Output

#	Time	Action
✓ 5	12:13:53	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000
✓ 6	12:14:21	SELECT * FROM orders WHERE ord_date='2012-09-10' LIMIT 0, 1000
✓ 7	12:14:51	SELECT * FROM orders WHERE salesman_id=5001 LIMIT 0, 1000
✓ 8	12:15:39	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000
✓ 9	12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
✓ 10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
✓ 11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000

MAX()

```

49 • SELECT AVG(purch_amt) AS average_order
50 FROM orders;
51
52 • SELECT MAX(purch_amt) AS highest_order
53 FROM orders;
54

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	highest_order			
▶	5760.00			

Result 8 ×

Output

Action Output

#	Time	Action
✓ 6	12:14:21	SELECT * FROM orders WHERE ord_date='2012-09-10' LIMIT 0, 1000
✓ 7	12:14:51	SELECT * FROM orders WHERE salesman_id=5001 LIMIT 0, 1000
✓ 8	12:15:39	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000
✓ 9	12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
✓ 10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
✓ 11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
✓ 12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000

MIN()

55 •
56
57

```
SELECT MIN(purch_amt) AS lowest_order
FROM orders;
```

Result Grid

lowest_order

65.26

Result 9

Output

Action Output

#	Time	Action
7	12:14:51	SELECT * FROM orders WHERE salesman_id=5001 LIMIT 0, 1000
8	12:15:39	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000
9	12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000

COUNT()

58 •
59
60

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Result Grid

total_orders

10

Result 10

Output

Action Output

#	Time	Action
8	12:15:39	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000
9	12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000

GROUP BY

```
60
61 • SELECT salesman_id,
62      SUM(purch_amt) AS total_sales
63 FROM orders
64 GROUP BY salesman_id;
65
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	salesman_id	total_sales
▶	5002	1349.45
	5001	8225.86
	5003	2590.90
	5005	270.65
	5006	1983.43

Result 11 ×

Output

Action Output

#	Time	Action
✓ 9	12:16:07	SELECT * FROM orders ORDER BY ord_date LIMIT 0, 1000
✓ 10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
✓ 11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
✓ 12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
✓ 13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000

```
66 • SELECT customer_id,
67      SUM(purch_amt) AS total_purchase
68 FROM orders
69 GROUP BY customer_id;
70
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	customer_id	total_purchase
▶	3005	1099.00
	3002	5825.26
	3009	2590.90
	3007	2400.60
	3001	270.65
	3004	1983.43

Result 12 ×

Output

Action Output

#	Time	Action
✓ 10	12:16:40	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000
✓ 11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
✓ 12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
✓ 13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 10...

GROUP BY + MAX()


```

71 • SELECT customer_id,
72       MAX(purch_amt) AS highest_purchase
73 FROM orders
74 GROUP BY customer_id;
75

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	customer_id	highest_purchase
▶	3005	948.50
	3002	5760.00
	3009	2480.40
	3007	2400.60
	3001	270.65
	3004	1983.43

Result 13 x

Output

Action Output

#	Time	Action
✓ 11	12:17:19	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000
✓ 12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
✓ 13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 10...
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0,...

GROUP BY + HAVING


```

76 • SELECT salesman_id,
77       SUM(purch_amt) AS total_sales
78 FROM orders
79 GROUP BY salesman_id
80 HAVING SUM(purch_amt) > 3000;
81

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	salesman_id	total_sales
▶	5001	8225.86

Result 14 ×

Output

Action Output

#	Time	Action
✓ 12	12:18:04	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000
✓ 13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(purch_amt) > 3000

```

82 • SELECT customer_id,
83       SUM(purch_amt) AS total_purchase
84 FROM orders
85 GROUP BY customer_id
86 HAVING SUM(purch_amt) > 2500;
87

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	customer_id	total_purchase
▶	3002	5825.26
	3009	2590.90

Result 15 ×

Output

Action Output

#	Time	Action
✓ 13	12:18:45	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 1000
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0, 1000
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(purch_amt) > 3000
✓ 19	12:23:25	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING SUM(purch_amt) > 2500

GROUP BY + HAVING + ORDER BY

```
87
88 • SELECT customer_id,
89       SUM(purch_amt) AS total_purchase
90 FROM orders
91 GROUP BY customer_id
92 HAVING SUM(purch_amt) > 1000
93 ORDER BY total_purchase DESC;
94
```

Result Grid

	customer_id	total_purchase
▶	3002	5825.26
	3009	2590.90
	3007	2400.60
	3004	1983.43
	3005	1099.00

Result 16

Output

Action Output

#	Time	Action
✓ 14	12:19:18	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 10...
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0,...
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(...
✓ 19	12:23:25	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 20	12:24:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...

BETWEEN + HAVING

```
95 • SELECT customer_id,
96       MAX(purch_amt) AS max_purchase
97 FROM orders
98 GROUP BY customer_id
99 HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
100
```

Result Grid

	customer_id	max_purchase
▶	3002	5760.00
	3009	2480.40
	3007	2400.60

Result 17

Output

Action Output

#	Time	Action
✓ 15	12:20:25	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1000
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 10...
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0,...
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(...
✓ 19	12:23:25	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 20	12:24:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 21	12:24:44	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVING M...

COUNT + HAVING

```
101 • SELECT salesman_id,  
102       COUNT(*) AS total_orders  
103 FROM orders  
104 GROUP BY salesman_id  
105 HAVING COUNT(*) >= 2;  
106
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	salesman_id	total_orders
▶	5002	3
	5001	3
	5003	2

Result 18 x

Output

Action Output

#	Time	Action
✓ 16	12:21:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT 0, 10...
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0,...
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(...
✓ 19	12:23:25	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 20	12:24:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 21	12:24:44	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVING M...
✓ 22	12:25:19	SELECT salesman_id, COUNT(*) AS total_orders FROM orders GROUP BY salesman_id HAVING COUNT(*) ...

MAX + GROUP BY + HAVING

```
107 • SELECT ord_date,  
108       MAX(purch_amt) AS highest_purchase  
109 FROM orders  
110 GROUP BY ord_date  
111 HAVING MAX(purch_amt) > 2000;  
112
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	ord_date	highest_purchase
▶	2012-10-10	2480.40
	2012-07-27	2400.60
	2012-09-10	5760.00

Result 19 x

Output

Action Output

#	Time	Action
✓ 17	12:22:03	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT 0,...
✓ 18	12:22:47	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING SUM(...
✓ 19	12:23:25	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 20	12:24:05	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING S...
✓ 21	12:24:44	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVING M...
✓ 22	12:25:19	SELECT salesman_id, COUNT(*) AS total_orders FROM orders GROUP BY salesman_id HAVING COUNT(*) ...
✓ 23	12:25:54	SELECT ord_date, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY ord_date HAVING MAX(...